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A 17-year retrospective study of institutional results for eye plaque brachytherapy of uveal melanoma using ^{125}I, ^{103}Pd, and ^{131}Cs and historical perspective

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ABSTRACT

PURPOSE: To compare overall survival, local and distant failure rates, ocular toxicity, and vision preservation in patients treated with eye plaque brachytherapy at Tufts Medical Center with those in the published literature.

METHODS AND MATERIALS: Records were reviewed for 53 patients with the diagnosis of uveal melanoma treated with plaque brachytherapy at Tufts Medical Center over the past 17 years. American Joint Committee on Cancer staging (T1, T2, or T3) were 4, 39, and 10 patients, respectively. All the patients were treated using ^{125}I ($n = 37$), ^{103}Pd ($n = 5$), or ^{131}Cs ($n = 11$) to a dose of 85 Gy (documented as 100 Gy before 1996 for the same physical dose).

RESULTS: With a mean followup of 75 months, 38 of 53 patients were still alive. Five patients (all ^{125}I) developed liver metastases (9%) with no evidence of local failure. There were 10 definitive local failures and four additional transpupillary thermo-therapy procedures performed to ensure local control for lesions slow to respond. Twelve patients (23%) required enucleation. At most recent followup, 32 patients (71%) maintained 20/200 vision or better in the treated eye. In this first report of ^{131}Cs plaque therapy with a mean followup of 20 months, there were two transpupillary thermo-therapy procedures and one definitive failure requiring enucleation after 10 months.

CONCLUSIONS: Our disease control and ocular results were comparable to those in the literature given the extended followup. We are developing a multi-institutional, prospective clinical protocol for considering radionuclide selection and other prescriptive criteria.

Keywords: Uveal melanoma; Eye plaque; Brachytherapy; ^{125}I; ^{103}Pd; ^{131}Cs

Introduction

When compared with enucleation, eye plaque brachytherapy provides adequate control of primary uveal melanoma tumors, superior vision, and globe preservation (1−5). Brachytherapy, when compared to external beam radiotherapy techniques, allows for dose limitation to the retina, optic nerve, lacrimal gland, and eyelids. Since the 2001 publication of the Collaborative Ocular Melanoma Study (COMS) on 657 patients, many institutions have published their experiences. These studies primarily include the use of two radionuclides: iodine-125 (^{125}I) (6−17) and palladium-103 (^{103}Pd) (18−21). The literature also contains some reports of cobalt-60 (^{60}Co) (22), iridium-192 (^{192}Ir) (22, 23), and ruthenium-106 (^{106}Ru) (20, 24−30).

In 2003, the American Brachytherapy Society (ABS) published recommendations for the treatment of uveal melanoma using brachytherapy plaques (31). These recommendations covered appropriate patient selection, plaque design,

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treatment planning, and dose prescription and delivery. Radionuclides available for eye plaque brachytherapy were reviewed, but specific recommendations regarding radionuclide selection were not made. Existing literature contains limited results examining radionuclide choice.

In this article, results from our institutional experience using ^{125}I, ^{103}Pd, and uniquely cesium-131 (^{131}Cs) were analyzed, focusing on overall survival, local control, distant metastases, ocular toxicity, and visual acuity (32). Implicit in this analysis, we compared outcomes as a function of radionuclide. Also provided is a comprehensive literature review covering publications spanning the last two decades, and comparison of these results with our observations.

Methods and materials

Records were available for 53 patients treated at Tufts Medical Center (TMC) between January 29, 1992 and July 14, 2009. Patients were included in this retrospective study if they met the following criteria: at least 4 years followup postimplant, followup until the time of enucleation or death, or followup within 6 months of the study end date (March 22, 2010). Table 1 summarizes patient characteristics. American Joint Committee on Cancer (AJCC) staging (T1, T2, or T3) were 4, 39, and 10 patients, respectively. Mean apical lesion height was 5.8 mm for the entire cohort and 6.3, 2.7, and 5.4 mm for the ^{125}I ($n = 37$), ^{103}Pd ($n = 5$), and ^{131}Cs ($n = 11$) patients, respectively. Table 2 presents the radiologic characteristics of each of these radionuclides.

Table 1
Patient characteristics by radionuclide (^{125}I, ^{103}Pd, and ^{131}Cs) and for the entire cohort

Radionuclide	^{125}I	^{103}Pd	^{131}Cs	Total
Number of patients	37	5	11	53
Age (yr)				
Mean	62	63	56	61
Range	39−85	58−70	32−79	32−85
Gender				
Men	22	2	5	29
Women	15	3	6	24
Affected eye				
OD	22	4	3	29
OS	15	1	8	14
T stage				
T1 (COMS-I)	1	1	2	4
T2 (COMS-II)	30	4	5	39
T3 (COMS-III)	6	0	4	10
Mean apical tumor height (mm)	6.3	2.7	5.4	5.8
Range	2.0−10.3	2.3−3.3	2.1−9.4	2.0−10.3
Mean basal tumor length (mm)	12.3	10.6	12.6	12.2
Range	4.5−18.8	8.9−13.2	9.3−18.2	4.5−18.8
Treatment era				
1992−1999	18	0	0	18
2000−2009	19	5	11	35
Length of implant (d)				
7	34	2	0	36
5	3	3	11	17

OD = right eye; OS = left eye; COMS = Collaborative Ocular Melanoma Study.

All the patients were seen for initial consult at the New England Eye Center of TMC for history and physical examination. Ophthalmologic examination included dilated fundoscopic examination to determine lesion location and ultrasound A and B scans to determine lesion size. Systemic workup, most often composing liver function tests and CT scan of the chest and abdomen, was conducted to evaluate for synchronous metastatic disease. Referral was made to radiation oncology for patients for whom plaque radiotherapy was considered a treatment option. For patients who were lost to followup within the TMC system, the primary ophthalmologist or primary care physician was contacted.

Surgical implantation

Brachytherapy plaques were implanted by the ophthalmologist (JSD) under conscious sedation or general anesthesia. Extraocular muscles were immobilized and transillumination was used for tumor localization. A dummy plaque (clear plastic disk) was placed to determine plaque position. Radioactive plaque placement was confirmed by a radiation oncologist. Treatment lasted 7 days (all treatments before November 2007) or 5 days (all treatments in November 2007 and thereafter). All the patients were treated after having signed informed consent. Patient records were reviewed as per protocol #6797 approved by the TMC Institutional Review Board.

Brachytherapy dosimetry

Patients were treated to a total physical dose of 85 Gy (documented as 100 Gy before November 1996) prescribed to a height of 5 mm from the tumor base for tumors $\leq$5 mm in height or to the tumor apex for tumors $>$5 mm in height. The prescribed dose follows the recommendations set forth by the ABS (31).

For each treatment, the appropriate source strength was ordered to deliver 85 Gy over our standardized implant duration (7 days before November 2007 and 5 days thereafter). The most influential factor for radionuclide selection was availability of the appropriate source strength to deliver

Table 2
Radiologic characteristics of ^{125}I, ^{103}Pd, and ^{131}Cs eye plaque implants

Radionuclide	^{125}I	^{103}Pd	^{131}Cs
Average photon energy (MeV)	0.028	0.021	0.030
Half-life (d)	59.4	17.0	9.7
Decay constant (h^{-1})	0.0005	0.0017	0.0030
Initial dose rate (Gy/h) @ 5-mm depth			
7-d implant	0.53	0.58	0.64
5-d implant	0.73	0.78	0.84

the prescribed dose over our standardized implant durations. For two specific cases, ^{103}Pd was selected in early 2007 because there was clinical reason to believe the tumor was fast growing. ^{131}Cs sources were first used in December 2007 because of their more uniform dose distribution when compared with ^{125}I and ^{103}Pd, and shorter half-life because of radiologic waste concerns. Most recently, ^{103}Pd has been used for thin lesions measuring less than 3.5 mm in apical height because of lower critical structure doses (33).

Plaque sizes ranged from 14 to 22 mm (chosen to accommodate the maximum tumor basal dimension with a 2–3-mm margin), and were either standardized COMS plaques or notched (about 23% of patients) with a lip for positioning and shielding of the optic nerve. Treatment planning was performed using the American Association of Physicists in Medicine TG-43 2D formalism (34, 35) for the model 6711 ^{125}I (Oncura, Inc., Princeton, NJ), model 200 ^{103}Pd (Theragenics, Inc., Buford, GA), and model CS-1 Rev2 ^{131}Cs (IsoRay Medical, Inc., Richland, WA) sources.

Followup

Followup time was measured starting from the day of implant. All patients were seen for first followup within 1 week of plaque removal to assess for complications. Second followup was scheduled for 1 month after plaque removal to assess for short-term side effects. To monitor lesion size and to assess long-term ocular toxicity, ultrasound B scans were obtained after explant with typical frequencies of every 3 months in the first 2 years, every 6 months in the third year, and annually thereafter. Local failure was defined per the COMS definition: (*1*) initial increase of at least 15% in height as measured by ultrasound, and (*2*) an additional 15% or 0.25 mm expansion of any tumor dimension observed on subsequent examination. However, these criteria may be too stringent based on current quality control test procedures and measurement variations because of reproducibility, user dependence, and equipment fluctuation (36).

Statistical methods

Time to local failure and death were calculated using the Kaplan–Meier product–limit method for the entire cohort. Univariate analysis was used to assess relationships between local failure and age (<60, ≥60 years), gender, plaque size—a surrogate for basal diameter—(<20, ≥20 mm), apical height (≤4.5, >4.5 mm), and implant duration. Patient and tumor characteristics analyzed were among the prognostic factors identified in COMS Report No. 28 (5). Using stepwise regression, multivariate models were developed to examine the relationship between local failure and gender, plaque size, and apical height.

Literature review

The literature (MEDLINE, Bethesda, MD) was reviewed to identify publications related to the use of plaque brachytherapy for the treatment of uveal melanoma. Initial search terms included brachytherapy, uveal melanoma, eye plaque, COMS, choroidal melanoma, and plaque brachytherapy. Thirteen relevant clinical reports were initially identified; the references of these papers were reviewed to identify additional sources. Prospectively gathered data (1, 18) and retrospective data were reviewed and analyzed. Regarding analysis of local control rates for the present cohort as compared with local control rates in the literature, direct comparison was made only between observed control rates; actuarial rates were reported but not analyzed.

Results

Overall survival

At a mean followup of 75 months (range, 2–214 months), 38 of 53 patients were still alive. Cause of death remained unknown in most of the patients who had passed away; however, 4 of 15 patients (27%) were known to have died from metastatic disease to the liver. Median survival was 107 months as illustrated in Fig. 1.

Local control

There were 10 definitive local failures (19%) with a mean time to failure of 30 months (range, 7–150 months). Most of the local failures occurred in men (90%), each of these resulting in enucleation. Likewise, most of the failures occurred with 20 mm or larger plaques

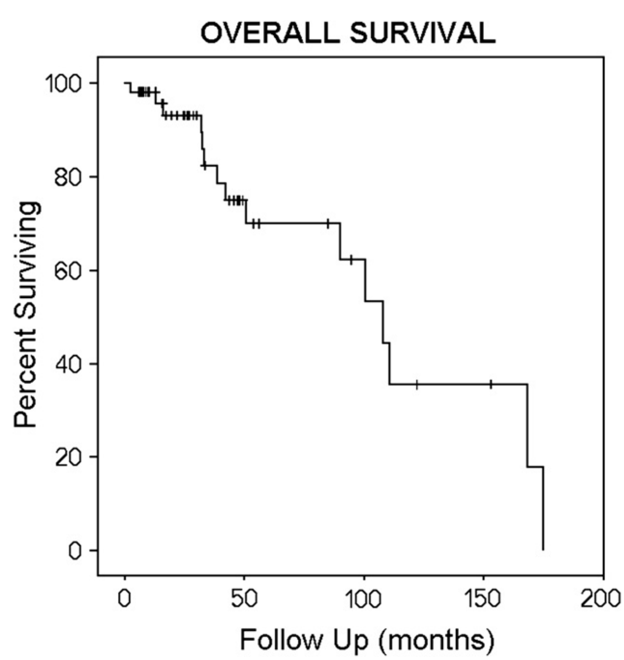

Fig. 1. Overall survival of the entire cohort ($n = 53$).

(90%), or before November 2007 with 7-day implants (90%). In addition, four transpupillary thermo-therapy (TTT) procedures were performed to ensure local control for lesions slow to respond (one ^{125}I, one ^{103}Pd, and two ^{131}Cs) in an attempt toward vision preservation. Time until local failure is presented by tumor stage at diagnosis in Fig. 2 and by radionuclide in Fig. 3. Results of the univariate analysis and stepwise regression analysis are presented in Table 3. On univariate analysis, gender ($p = 0.01$), plaque size ($p = 0.01$), and apical height ($p = 0.02$), were predictive of local failure. Gender ($p = 0.03$) and plaque size ($p = 0.03$) remained predictive on multivariate analysis.

Distant failure

Five patients (all ^{125}I and AJCC stage T2) developed liver metastases (9%) with no evidence of local failure at a mean time to failure of 19 months (range, 12–28 months). All systemic failures occurred in patients aged 48–63 years, and in patients treated with 7-day implants.

Ocular toxicity

Cataract development was the most common treatment-related ocular toxicity. At study completion, 35 patients had visually significant cataracts with 12 of these patients (34%) having documented cataracts at treatment initiation. Twenty-eight (80%) of these cataracts occurred in patients aged 50 years or older, with 26 cataracts in patients with T2 disease. Eighteen cataracts (51%) became evident within the first year after treatment and an additional nine (26%) in the second year. A total of 10 cataract surgeries were performed with two in patients having preexisting cataracts that progressed after treatment. Eight of the 10 required cataract surgeries were performed in patients with T2 disease. Twelve patients (23%) required enucleation (11 ^{125}I and 1 ^{131}Cs): 10 for disease control and 2 for

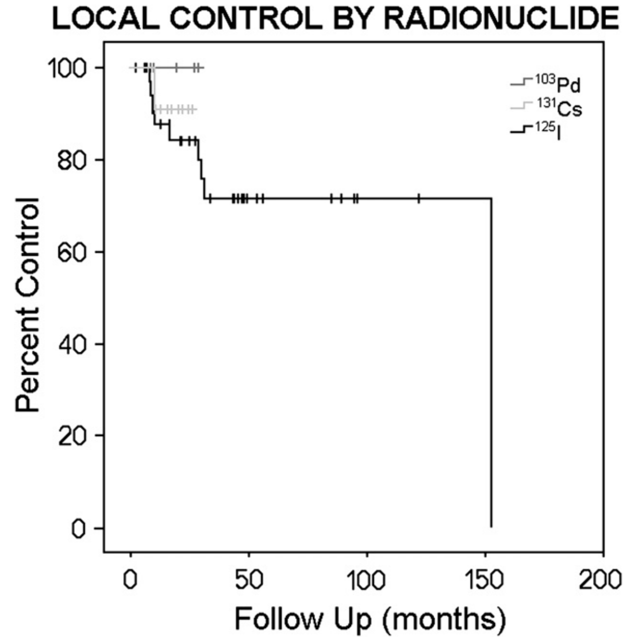

Fig. 3. Local control by radionuclide ^{103}Pd, ^{131}Cs, and ^{125}I after the vertical order of the legend.

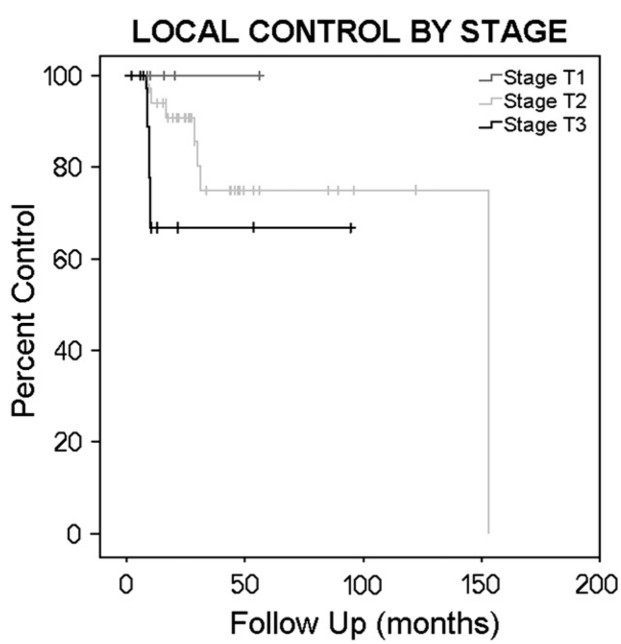

Fig. 2. Local control by American Joint Committee on Cancer T stage T1, T2, and T3 after the vertical order of the legend.

Table 3
Results of univariate and multivariate analyses for the entire cohort

Radionuclide	Univariate analysis	Multivariate analysis
Number of patients		
Age (yr)		
<60	$p > 0.05$	N/A
≥60		
Gender		
Men	$p = 0.01$	$p = 0.03$
Women		
Radionuclide		
^{125}I	$p > 0.05$	N/A
^{103}Pd		
^{131}Cs		
Apical tumor height (mm)		
≤4.5	$p = 0.02$	$p > 0.05$
>4.5		
Plaque size (mm)		
<20	$p = 0.02$	$p = 0.03$
≥20		
Duration of implant (d)		
7	$p > 0.05$	N/A
5		

N/A = not applicable (factor removed from multivariate analysis based on nonsignificance in univariate model).

symptom management. Eleven of these 12 enucleations occurred in men. Symptom management included eight TTT for exudative retinal detachment on eight patients. Table 4 outlines ocular toxicity data including cataract incidence, retinal detachment, and radiation retinopathy/maculopathy in the literature and in the present cohort.

Vision preservation

At most recent followup, 32 patients maintained 20/200 vision or better in the treated eye. Table 5 outlines vision preservation in the literature and present cohort. Vision was preserved less frequently for lesions with apical height greater than 5 mm and for those treated with plaques 20 mm or larger.

^{131}Cs results

In this first report of ^{131}Cs eye plaque brachytherapy for 11 patients with a mean followup of 20 months, there was one definitive local failure requiring enucleation after 10 months and 2 patients (18%) requiring TTT procedures (both 7 months after treatment) because of slow-responding disease. There were no observations of distant failure. The rates of enucleation and cataract formation were 9% and 62%, respectively. Retinopathy and retinal detachment rates were each 18%. There were no observations of vitreous hemorrhage or keratitis. For visual acuity, 9 of 11 (82%) ^{131}Cs patients maintained 20/200 vision or better, and 8 of 11 (73%) lost ≤2 lines on the Snellen chart.

Discussion

Literature review

Review of the literature revealed several COMS reports presenting data from the only randomized multi-institutional prospective trial of eye plaque brachytherapy for the treatment of uveal melanoma (1, 3–5). Four reports of a single-center prospective trial using ^{103}Pd have been published (18–21), and a number of retrospective reviews have been published. Table 6 outlines the number of patients, followup, radionuclide selection, local control rates, distant failure rates, and overall survival rates in each of these studies. Ocular toxicity data is presented in Table 4 and vision preservation data in Table 5.

Authors of each study reported outcomes in the format most appropriate to their data. Local control, distant failure rates, and overall survival rates were usually reported in

Table 4
Toxicity rates from the literature and the TMC cohort

Nuclide	Study	Enucleation (%)	Cataract development (%)	Radiation maculopathy or retinopathy (%)	Vitreous hemorrhage (%)	Retinal detachment (%)	Keratitis (%)
^{106}Ru	Heindl *et al.* (25)	14	—	13(R)	—	—	—
	Gunduz *et al.* (27)	11	35	40(M)	—	—	—
	Lommatzsch (29)	20.7	2	26.8(M)	3	—	—
	Literature average	15.2	18.5	33.4(M) 13(R)	N/A	N/A	N/A
^{125}I	COMS (1–5)	13	—	—	—	—	—
	Correa *et al.* (6)	10	31.6	4.7(M) 7.5(R)	3.3	4.1	—
	El-Ghamry *et al.* (7)	—	31	27(R)	3	—	—
	Jensen *et al.* (8)	3	19	13(M) 7.5(R)	15	—	—
	Lumbroso-Le Rouic *et al.* (9)	1.5	50.3	18.3(M)	3.4	3	2.8
	Nag *et al.* (10)	—	—	40(R)	—	—	—
	Jones *et al.* (11)	11	—	62.8(R)	—	—	—
	Quivey *et al.* (12)	—	14	18.2(M)	20.8	—	—
	Fontanesi *et al.* (13)	10	30	21.5(R)	—	—	—
	Packer *et al.* (15)	17.2	45.3	23.4(R)	21.9	—	—
	Petrovich *et al.* (16)	15	—	—	—	—	—
	Literature average	10.1	31.6	13.6(M) 6.4(R)	11.2	3.6	N/A
^{103}Pd	Finger *et al.* (18)	8	58	—	—	—	—
	Finger *et al.* (19)	3.5	23	14(M)	—	—	—
	Literature average	5.8	40.5	N/A	N/A	N/A	N/A
Other	DePotter *et al.* (22)	22	—	87(R)	—	—	—
	Shields *et al.* (24)	24	66	24(M) 25(R)	23	—	—
	Literature average	23	N/A	24(M) 56(R)	N/A	N/A	N/A
	TMC total	22.6	56	25(R)	7.5	26	3.6

TMC = Tufts Medical Center; M = radiation maculopathy; R = radiation retinopathy; COMS = Collaborative Ocular Melanoma Study.

Table 5
Visual acuity data from the literature and the Tufts Medical Center cohort

Nuclide	Study	Visual acuity retention (%)
^{125}I	COMS (1–5)	20/200, 57(3 yr)
	Fontanesi *et al.* (13)	20/200, 41(4 yr)
	Packer *et al.* (15)	40/200, 45(5 yr)
^{103}Pd	Finger *et al.* (19)	20/200, 79(5 yr), 69(10 yr)
	Finger *et al.* (20)	20/200, 73(5 yr)
	Finger (21)	<3 lines, 62(3 yr)
Other	DePotter *et al.* (22)	<3 lines loss, 36(5 yr)
	TMC total	20/200, 51

actuarial terms and therefore precluded meta-analysis without access to the complete data sets. Further, ocular toxicity and vision preservation data were reported in various metrics, which complicated comparisons among the studies.

As evidenced by the literature review, there is extensive experience treating uveal melanoma with plaque brachytherapy (mostly with ^{125}I and ^{103}Pd). As described by Correa *et al.* (6), direct comparisons between individual series become quite difficult because of variations in patient populations, tumor stage, plaque type, and radionuclide used. In most series, the median tumor stage is T2. However, as was the case with the Spanish cohort, the present study includes an unusually high proportion (19%) of T3 tumors—especially given our use of Gamma Knife ^{60}Co stereotactic radiosurgery in 2000 as an alternative treatment modality for large lesions or those too close to the optic disc to be adequately treated using plaques. Moreover, the mean apical height of melanomatous lesions was among the highest reported at 5.8 mm, suggesting a TMC patient population with more advanced disease at presentation, perhaps because of its role as a major referral center. The observed local control rate of this study was 81%, with 9 of 10 definitive local failures occurring in the ^{125}I arm of the cohort. The average local control rate from the literature was 87%.

Treatment of more locally advanced tumors is likely also reflected in the noticeably high enucleation rate of 23%. Additionally, 11 of 12 enucleations in our cohort took place in the ^{125}I arm, indicating a 30% enucleation rate for this radionuclide at TMC; this is in contrast to the literature average ^{125}I enucleation rate of about 10%. The enucleation rate in the present cohort likely also reflects the selection bias inherent to retrospective studies. Of all the patients treated at TMC with eye plaque brachytherapy, only those with adequate followup were included in this analysis. Patients undergoing enucleation were more likely to followup within TMC where most of the enucleations were performed.

Five-year local control rates (many of them actuarial) in the literature ranged from 85% to 97%; however, local control with followup beyond 5 years is limited. Although most of the local failures occurred within 3 years, the present study, with followup spanning 17 years, demonstrated that local failures might continue beyond 5 years. One patient underwent enucleation 12 years postimplant because of vision loss secondary to nonresolving retinal hemorrhage and progressive painful dry eye. The enucleation specimen contained residual melanoma on pathologic review. Similarly, Char *et al.* (17) have shown that approximately 2% of patients treated with ^{125}I plaques may fail between 5 and 15 years.

As expected, apical height and plaque size (a surrogate for basal diameter) predicted for local failure on univariate analysis, with superior local control for smaller lesions. More definitive failures occurred with plaques 20 mm or greater. On multivariate analysis, apical height did not significantly predict for local failure but plaque size did. Basal diameter appeared to be more closely related to local control than did apical height. Similarly, maximum basal tumor diameter was identified as a significant predictor of death in the COMS series (5).

Although not significant on stepwise regression analysis, improved local control was seen with 5-day vs. 7-day implants, which might reflect the more modern treatment era (5-day implants occurred since November 2007) or possibly a radiobiologic advantage of higher dose rate. Indeed, biologically effective dose increases with increasing dose rate when calculated using the equation for temporary brachytherapy implants proposed by Dale and Jones (37).

Interestingly, both local failures and enucleations occurred much more frequently in males than in females. As we were unable to find a precedent in the literature, it remains unclear why gender predicted so strongly for local failure in our cohort. However, similar findings have been reported elsewhere; treatment failure occurred equally in 10% of men and of women in the COMS cohort, but 17% of men and 8% of women underwent enucleation (1).

Excellent local control rates seen in the 5 patients treated with ^{103}Pd likely reflect the small size of these lesions (mean apical height = 2.7 mm; range, 2.3–3.3 mm). Moreover, a total dose of 85 Gy was prescribed to 5.0 mm, at least 1.7 mm beyond the lesion height. Following this prescriptive paradigm, the apex of each of the five lesions was treated with a dose larger than the prescribed dose.

In the present cohort, 1 patient treated with ^{125}I, 1 patient treated with ^{103}Pd, and 2 patients treated with ^{131}Cs had lesions that were slow to respond and that increased slightly in apical height initially after plaque implantation. Two of these lesions (one ^{103}Pd and one ^{131}Cs) met the first of the two-tiered COMS criteria for local relapse (an increase in apical height of at least 15%). All the 4 patients were treated with TTT and lesions regressed; thus none met the secondary COMS criteria for local relapse. The first such TTT procedure was performed in December 2008. Had it been previously available, it is unclear whether this procedure could have successfully salvaged some local failures occurring in earlier years.

Table 6
Clinical data including patient characteristics, followup, and outcomes from the literature review and the TMC cohort

Nuclide	Study	No. of patients	Mean dose (Gy)	Mean followup (mo)	Mean initial lesion dimensions	Local control (%)	Distant failure (%)	Overall survival (%)
^{106}Ru	Heindl *et al.* (25)	100	—	—	—	93(5 yr)	10(5 yr)	—
	Damato *et al.* (26)	458	115	46.8[a]	10.6/3.2 (D/T)[a]	99(2 yr), 98(5 yr), 97(7 yr)	—	—
	Gunduz *et al.* (27)	630	—	62.4	(D) < 10, 4 (T)[a]	91(5 yr)	12(5 yr), 22(10 yr)	—
	Seregard *et al.* (28)	266	100	43	10/4.4 (D/T)	90	11(4 yr), 14(5 yr)	83.1
	Lommatzsch (29)	309	100	80	COMS-I (100%)	84(5 yr), 69.9(6.7 yr)	12.9(7 yr)	87.1
	Literature average	353	105	61.8	10.3/3.9 (D/T)	90, 91.5(5 yr)	12(5 yr)	85.1
^{125}I	COMS (1–5)	657	85	60	11.4/4.8 (D/T)	90	10(5 yr), 18(10 yr)	82(5 yr)
	Correa *et al.* (6)	120	83	50.4[a]	12.2/5.9 (D/T), COMS-II (73%)	95.3(2 yr), 88.4(5 yr)	9.5(2 yr), 20.3(5 yr)	86.7, 94.4(2 yr), 84.1 (5 yr)
	El-Ghamry *et al.* (7)	119	113	67	10.6/4.2 (D/T)	97.5	8.3	75(5 yr)
	Jensen *et al.* (8)	156	—	74.4[a]	—	92(5 yr)	10(5 yr), 27(10 yr)	80, 83(5 yr)
	Lumbroso-Le Rouic *et al.* (9)	136	112	62[a]	10.3/4.7 (D/T)	98.5	5.9, 1.7(2 yr), 4(5 yr)	89, 93.8(2 yr), 88.3(5 yr)
	Nag *et al.* (10)	78	85–100	49[a]	COMS-II (100%)	93(5 yr)	—	86(5 yr)
	Jones *et al.* (11)	63	85–100	36	4.5 (T)	84	4.8(3 yr)	—
	Quivey *et al.* (12)	150	95	68	9.7/3.7 (D/T)	81(5 yr), 78(5.6 yr)	17.3(5 yr)	78
	Fontanesi *et al.* (13)	144	79	46	975 (V)	97.7	5.5(4 yr)	—
	Quivey *et al.* (14)	239	70	36	10.9/5.5 (D/T)	91.7(3 yr), 82(5 yr)	7.5(3 yr), 12(5 yr)	—
	Packer *et al.* (15)	64	91	64	—	92.2(5 yr), 87.5(5.3 yr)	15.6(5 yr), 17.2(6 yr)	82.8
	Petrovich *et al.* (16)	85	102.6	37	6.1 (T), T2/T3 (96%), COMS-II (100%)	—	11, 10.6(8 yr)	88, 88(5 yr), 84(8 yr)
	Char *et al.* (17)	449	70	—	4.6 (T)	87	—	—
	Literature average	168	92	52 59[a]	11.55/5.5 (D/T), 10.5/4.35 (D/T)	93.5, 88.1(5 yr)	8.4, 3.7(2 yr), 6.2(3 yr), 12.7(5 yr), 22.5(10 yr)	84, 94.1(2 yr), 83.8(5 yr)
	TMC	37	85	100	12.3/6.3 (D/T), T2 (81%)	76	13.5	59.5
^{103}Pd	Finger *et al.* (18)	24	84.1	35.5 30[a]	2.6 (T)	95.9	4.1(6 yr)	91.7
	Finger *et al.* (19)	400	73	51	T1/T2 (85.5%), COMS-II (92%)	96.7	6, 3.3(2 yr), 7.4(5 yr), 13.4(10 yr)	87
	Literature average	400	73	43.3	N/A	96.3	N/A	87
	TMC	5	85	22.5	10.6/2.7 (D/T), T2(80%)	100	0	100
^{131}Cs	TMC	11	85	20	12.6/5.4 (D/T), T2/T3 (82%)	91	0	100
Other	DePotter *et al.* (22)[b]	93		78	3 < (T)[a] < 6	84(5 yr), 73(10 yr)	12(4 yr), 13(5 yr), 16(10 yr)	81
	Lean *et al.* (23)[c]	56	94.5	39	13.2/6.8 (D/T), T3 (68%)	91	9(4 yr)	—
	Shields *et al.* (24)[d]	354	—	—	(T) > 8	91(5 yr), 87(10 yr)	30(5 yr), 55(10 yr)	—
	Literature average	168	N/A	59	N/A	91, 87.5(5 yr), 80(10 yr)	10.5(4 yr), 21.5(5 yr), 35.5(10 yr)	N/A

D = diameter; T = thickness; V = volume (cm^3); TMC = Tufts Medical Center; COMS = Collaborative Ocular Melanoma Study; N/A = not applicable.

[a] Median value.

[b] ^{125}I ($n = 63$), ^{192}Ir ($n = 15$), ^{60}Co ($n = 12$), and ^{106}Ru ($n = 3$).

[c] ^{125}I and ^{192}Ir.

[d] ^{125}I and ^{106}Ru.

Most of the literature reports overall survival rates of about 85%; the overall survival in the present study again is slightly lower, at about 72%. All the patient deaths in this study were in the ^{125}I arm with a mean followup time of ~100 months (range, 8−214 months), about twice as long as the literature average; a lower overall survival rate is thus expected given such extended followup. Establishment of a standardized time metric such as 5-year survival would facilitate study comparisons. Examination of our 5-year survival rate of 83% shows good agreement when compared with the COMS followup time. Unfortunately, cause of death remained unknown for most of the patients who died in the present cohort but only four deaths were known to be directly related to metastatic uveal melanoma.

Cataract development rate for our study, about 56%, also tended to be higher than the literature at 35−40%; certainly the high percentage (62%) of cataract developments in the ^{131}Cs arm boosted this number, but the high incidence of cataract formation may also be because of the longer followup time in our study.

Radiation-induced maculopathy/retinopathy rates (25%) for the present study are similar to the 20% observed by Shields *et al.* (24) whose retrospective study also involved multiple radionuclides. Both these results also compare favorably with the maculopathy/retinopathy results (32%) from the literature at large (not segregated by radionuclide). Most of the cases of radiation retinopathy (10 of 13 cases) in the present study occurred between 1 and 3 years postimplant.

There is less literature available regarding postoperative vitreous hemorrhage rates, with most based on experience only with ^{125}I. Still, the TMC results of 7.5% are slightly better than the 11.2% of the literature at large. Even less information was present in the literature regarding rates for retinal detachment and keratitis; our retinal detachment rate of 26% seems high, whereas our keratitis rate of 3.6% is comparable with studies such as Lumbroso-Le Rouic *et al.* (9) who report a similar value of about 2.8%.

The ideal radionuclide(s) for use in eye plaque brachytherapy is not known. Published reports include experiences using ^{125}I, ^{103}Pd, ^{106}Ru/^{106}Rh, ^{192}Ir, and ^{60}Co. To our knowledge, this article is the first to report the use of ^{131}Cs in plaque brachytherapy for uveal melanoma. With an average photon energy of 30 keV and a half-life of 9.7 days, ^{131}Cs provides a more homogenous dose distribution than ^{103}Pd or ^{125}I (38), and higher initial dose rate. Such dosimetric characteristics may make ^{131}Cs a desirable radionuclide for use in plaque brachytherapy for large uveal melanomas. However, it is unclear whether or not these physical properties lead to clinically significant outcomes. Direct comparisons of outcomes by radionuclide have not yet been published to our knowledge. This study aims to present preliminary data comparing ^{125}I, ^{103}Pd, and ^{131}Cs. Dosimetric evaluations of these three radionuclides using Monte Carlo methods for radiation transport and conventional treatment planning systems have been performed (39, 40). Results indicate that radionuclide selection as a function of lesion height and applicator size produce predictable alterations in dosimetry. On the basis of our observations, this does not seem to alter clinical outcomes. Univariate analyses indicate that radionuclide selection does not predict for local failure. The current retrospective analysis suggests that these three radionuclides provide similar local control. However, this hypothesis has yet to be tested in the form of a randomized controlled trial. This study serves as preliminary results for a prospective clinical trial, comparing effectiveness and ocular toxicity of ^{125}I, ^{103}Pd, and ^{131}Cs-laden plaques for the treatment of uveal melanoma.

Conclusion

The results of this retrospective study of 53 patients treated with eye plaque brachytherapy for uveal melanoma between 1992 and 2009 at TMC are consistent with those reported in the literature. Uniquely included are patients treated with ^{131}Cs where results were comparable to those for other radionuclides (i.e., ^{125}I and ^{103}Pd). This work supports development of a prospective clinical protocol for considering radionuclide selection and other prescriptive criteria.

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